

Family Chironomidae

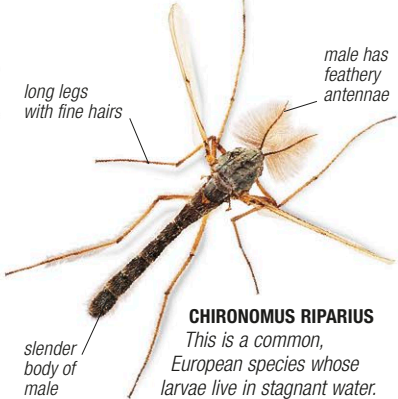
Non-biting midges

Occurrence 5,000 spp. worldwide; near ponds, lakes, and streams; swarms occur at dusk



These delicate, pale brown or greenish flies resemble mosquitoes, but do not have functional mouthparts. They are 1/32–1/32 in (1–9 mm) long. The males have a slender body and feathery antennae, while the females are stout, with hairy antennae. Most of the 2–3-year life cycle is spent as larvae; adults live no longer than 2 weeks. They mate on the wing in a swarm, and lay their eggs in a mass of

sticky jelly on water or plants. The larvae are plentiful in freshwater; up to 100,000 larvae have been recorded in 11 square ft (1 square meter).



Family Culicidae

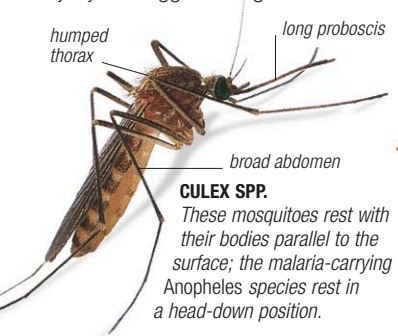
Mosquitoes

Occurrence 3,500 spp. worldwide, especially in warmer regions; everywhere near water

Habitat All

Mosquitoes are among the world's most dangerous pests, spreading malaria, yellow fever, and several other potentially fatal diseases. They are 1/8–3/4 in (0.3–2 cm) long, with narrow bodies, a humped thorax, and slender legs. Males often feed on flowers,

but females feed on blood—chiefly from mammals and birds—and have syringelike mouthparts for piercing skin. They lay their eggs on stagnant water.



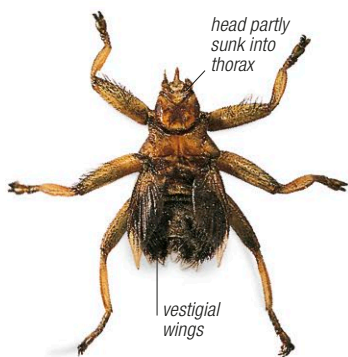
Family Hippoboscidae

Louse flies

Occurrence 200 spp. worldwide, mainly in tropical and subtropical regions; ectoparasites of birds and mammals



These dull brown, stout, flat, and hairy flies, mostly 5/32–9/32 in (4–7 mm) long, have a short proboscis and strong, clawed legs for gripping hairs or feathers. Some have fully formed wings, but in others the wings are vestigial, or absent altogether. Females lay mature larvae (prepupae) one at a time; these develop inside a “uterus,” feeding on secretions from special “milk” glands. Adult louse flies are blood-suckers.



CRATAERINA PALLIDA
Also called the swift louse fly, this European species is parasitic on swifts in their nests. Three-quarters of all louse fly species parasitize birds.

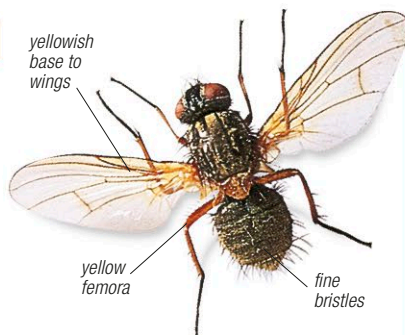
Family Muscidae

Muscids

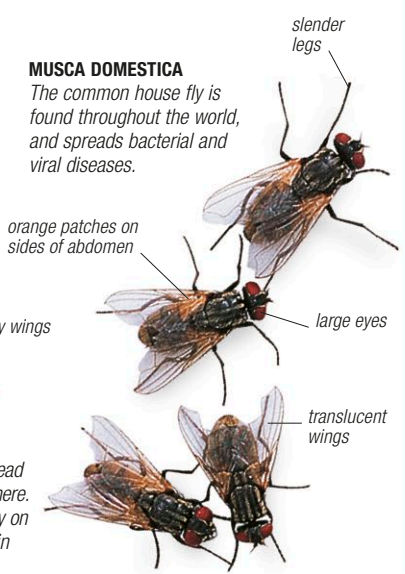
Occurrence 4,000 spp. worldwide; on flowers, dung, decaying matter, near mammalian hosts

Habitat Terrestrial

The muscid family includes one of the world's most successful insects—the common house fly—together with its many relatives. Muscids are usually drably colored, with bristly bodies up to 3/8 in (1.2 cm) long, and mouthparts that are spongelike for lapping fluids, or piercing for sucking blood. Female houseflies can lay 100–150 eggs per day in rotting matter, dung, fungi, water, or plants. Maggots grow fast and, in ideal conditions, pupate in just over a week. These flies can carry infections such as typhoid and cholera.



MYDAEA CORNI
This species, found in Europe and Asia, inhabits dung. The larvae are predators on small organisms inside the dung.



MUSCA DOMESTICA
The common house fly is found throughout the world, and spreads bacterial and viral diseases.

HELINA OBSCURATA
This species is widespread in the Northern Hemisphere. The larvae are predatory on small organisms found in soil and humus.

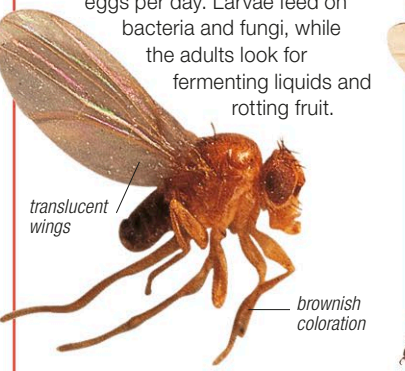
Family Drosophilidae

Pomace flies

Occurrence 2,900 spp. worldwide; near decaying vegetation, fruit, fungi, fermenting liquids

Habitat All

Also known as lesser fruit flies or vinegar flies, these small, yellow, brown to reddish brown, or black species have light or bright red eyes. The thorax and the abdomen may be striped or spotted. Pomace flies are 1/32–1/4 in (1–6 mm) long. Males make a buzzing sound with their wings to attract mates. Eggs are laid near a food source at the rate of 15–25 eggs per day. Larvae feed on bacteria and fungi, while the adults look for fermenting liquids and rotting fruit.



DROSOPHILA MELANOGASTER
The best-known pomace fly, the laboratory fruit fly is often used for genetics studies, as it breeds quickly and has large chromosomes in its salivary glands.

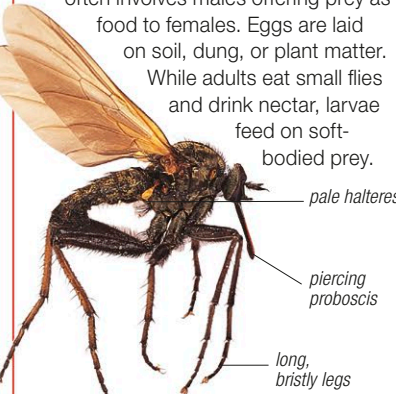
Family Empididae

Dance flies

Occurrence 3,000 spp. worldwide, mainly in the Northern Hemisphere; adults on tree trunks, larvae in rotten wood



The common name of this family comes from the dancing motions of mating swarms. Dance flies are 3/64–1/2 in (1.5–15 mm) long. Most species have a stout thorax, an elongated abdomen, a rounded head with large eyes, and a downward-pointing, sharp proboscis with which they can stab prey. Courtship often involves males offering prey as food to females. Eggs are laid on soil, dung, or plant matter. While adults eat small flies and drink nectar, larvae feed on soft-bodied prey.



EMPIS SP.
These flies inhabit damp vegetation. The long proboscis enables the fly to “stab” its prey and extract fluids on which it feeds.

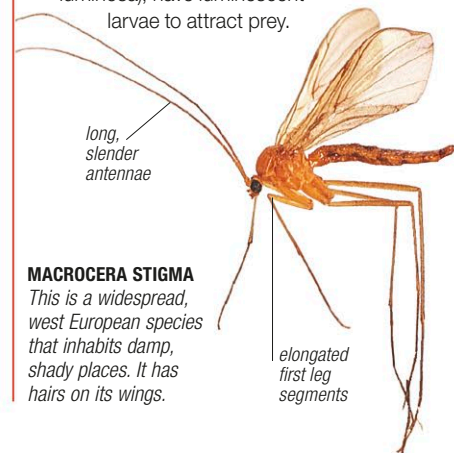
Family Keroplatidae

Fungus gnats

Occurrence 1,000 spp. worldwide; in forests, caves, and other moist places



Usually found in association with fungi in dark, damp forest habitats, keroplatid fungus gnats also occasionally live in caves. Adult keroplatids generally have dark bodies and may have patterned wings; one group mimic wasps. The larvae of some feed on fungi, but others are predators, ensnaring other insects and worms in slimy threads and killing them with oxalic acid produced by mouth glands. Several species, including the New Zealand cave “glowworm” (Arachnocampa luminosa), have luminescent larvae to attract prey.



MACROCERA STIGMA
This is a widespread, west European species that inhabits damp, shady places. It has hairs on its wings.